

Design, testing and analysis of a N_2O /paraffin wax hybrid rocket combustion system for low-cost propulsion applications

João Miguel Firmino Moreira
joao.f.moreira@tecnico.ulisboa.pt

Instituto Superior Técnico, Universidade de Lisboa, Portugal

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Abstract

The development of new viable propulsion systems that focus on low cost, simple throttling, restart and stop capabilities, and safety has been a major research focus in the aerospace industry. One standout solution is the hybrid rocket motor, which stores fuel and oxidizer separately in different phases. This configuration, using high regression rate fuels known as liquefying fuels, such as low-cost paraffin wax, combined with self-pressurizing oxidizers like N_2O , offer enhanced performance, a simple single-port design, and peak specific impulse over a wider range, being less sensitive to O/F shift during operation. To investigate the combustion of this promising propellant combination, a hybrid combustion setup with an optical access fuel slab burner was designed. The propellant performance tests included temporally resolved regression rate measurements using two distinct methods, one of which provided spatial resolution of the regression rate. These results also allowed for the spatial resolution of the flux-dependent regression rate law used in hybrid rocket design. Schlieren imaging and OH^* chemiluminescence techniques were applied to analyze the diffusion flame and thermal boundary layer, while color imaging revealed the liquid layer on fuel surface. Local regression rate exhibited a typical trend, decreasing after ignition region and then displaying a steadier behaviour. Both methods showed a low dependency on oxidizer flux. Mass fuel consumption calculated using both methods aligned and both demonstrated similar responsiveness to oxidizer flux. A lower position of the flame and boundary layer was identified with higher flux. Additionally, the flame's thickness increased with fuel length.

Keywords: Hybrid Rocket Motor, Paraffin, N_2O , Slab Burner, Regression Rate

1. Introduction

Hybrid rocket engines (HRE), unlike other chemical rockets, store the fuel and oxidizer separately in different phases. Figure 1 shows the typical configuration of hybrid rocket propulsion systems.

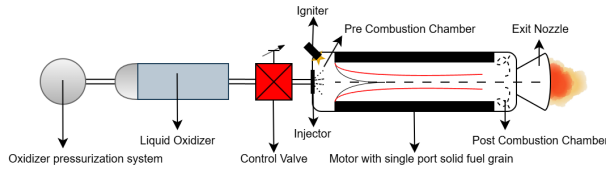


Figure 1: Hybrid Rocket Propulsion System.

This configuration requires only a single feed system line into the chamber containing the solid fuel grain. The oxidizer's delivery rate can be easily controlled using a main run valve. Furthermore, the fuel regression rate in these systems has been demonstrated to be independent of the pressure across most of the oxidizer flux range [1]. As a result, the thrust can be simply modulated by ad-

justing the rate of oxidizer delivery.

Additionally, using high regression rate fuels can enhance the performance of these rockets. When combined with a self-pressurizing oxidizer, such as nitrous oxide, the configuration becomes even simpler and reduces the additional weight needed for the pressurization system. Nitrous oxide is also less sensible to O/F shifts as the fuel burns and the port opens, allowing for peak performance for a wider range [2]. All these unique features make these systems simpler, more cost-effective, safer, and capable of throttling.

Historically, the performance of classic hybrid fuels was limited by the diffusion-evaporative nature of the combustion process, characterized by slow fuel evaporation and the blocking effect, as described by Marxman *et al.* [3]. The most effective and simpler solution to this issue emerged at the turn of the century with the discovery of a new class of high regression rate fuels, known as liquefying hybrid fuels.

According to the theory developed by Karabeyo-

glu *et al.* [4], during combustion, liquefying fuels form a very thin, low-viscosity, low-surface-tension liquid layer on the surface of the fuel grain. This layer becomes more unstable under the shear forces driven by the oxidizer gas flow in the port. For this class of fuels, the viscosity of the layer is so low that as instability amplifies, the thin layer creates non-linear roll waves and entrainment of droplets into the gas stream, greatly increasing the overall fuel mass transfer rate. Mixing is increased, and the fuels are no longer limited by diffusive heat transfer to the fuel from the combustion zone. For all these reasons, higher surface regression rates can be achieved than those achievable with conventional polymeric fuels that rely solely on evaporation. This theory is applicable to any material that form a liquid layer on its burning surface, including the homologous series of n-alkanes (C_nH_{2n+2}), with intermediate range of carbon numbers (16-50) corresponding to paraffin wax [5].

Experiments conducted at Stanford University using small-scale motor tests validated the predicted regression rates from liquid layer theory. The tests using paraffin based fuels report spatially averaged regression rates 3-4 times faster than burning rates of classic polymeric fuels [4].

The small-scale laboratory tests at Stanford University were then scaled up to chamber pressures and mass fluxes characteristic of operational systems well above critical conditions for paraffin wax [1]. The regression rate behavior observed in small-scale tests remained consistent under these operational conditions. These tests demonstrated weak dependency on the grain length. Additionally, it was confirmed that varying pressures do not significantly affect the regression rate. The results from these experiments validate the simplification of the original local regression rate predicted in classic theory by Marxman *et al.* [3] to equation 1:

$$\dot{r} = a_o G_{O_x}^n \quad (1)$$

On this expression, the regression rate depends only on the propellants combination given by the empirical constants (a_o and n), and the average oxidizer mass flux (G_{O_x}). Typically, the regression rate data fitted to this equation are averaged based on the total burn time and the total change in mass or height of the fuel grain measured before and after a hybrid motor burn test.

Several optical visualization experiments that provide full access to the hybrid combustion mechanisms, have been used to fully comprehend the combustion mechanisms of liquefying propellants. These experiments use small quantities of fuel in a slab geometry, and the combustion is usually limited to the top 2D surface of the fuel.

In the low flux regime, Nakagawa and Nikone [6]

observed the liquid layer entrainment mechanism described by Karabeyoglu *et al.* [4] using paraffin wax and gaseous oxygen, and demonstrated the inverse dependence of fuel viscosity on regression rate as predicted by Karabeyoglu *et al.* [4]. In more recent years, Jens [7] applied edge detection techniques to Schlieren images to identify the boundary-layer edge, and used peak OH* chemiluminescence intensities to locate the flame. The quantification of these mechanisms was conducted at different pressures and with both classic and liquefying fuels. The tests included blackened paraffin wax, as used in this work, at both atmospheric and supercritical pressure.

Optical visualization experiments using slab burners have also been used for spatially and temporally resolved regression rate measurements at atmospheric pressure in low oxidizer flux regime, using GOX and paraffin wax propellants. Dunn *et al.* [8] reported temporally and spatially resolved regression rate with sub-pixel accuracy for the middle portion (28-78%) of a 73 mm long fuel slab. Within this portion, the regression rate was spatially averaged and used to determine the constants of the hybrid regression rate law presented in equation 1. The results demonstrated great agreement with the averaged results reported by Nakagawa and Nikone [6]. More recently, Surina *et al.* [9] evaluated the accuracy of different image processing techniques to automate fuel mask segmentation for tracking fuel edges. The spatially averaged regression rate for different fluxes tended to overestimated the results from other slab tests. However, the results using these methods were consistent with manual regression rate measurements and spatially averaged results from sub-scaled hybrid motor firing tests.

As described so far, GOX has predominantly been the oxidizer of choice in slab burner experiments. Clearly there is a need for a hybrid combustion visualization experiment featuring N_2O , which offers unique features and broad potential applications in aerospace market from satellite micro-launchers to space exploration. Moreover, the few existing slab experiments that spatially resolve the regression rate consistently average the local rates to fit them into the hybrid fuels regression law assuming constant steady behaviour across the fuel length. The local regression rate has only been reported for smaller segments of the fuel length where the rate is more stable. Considering all these factors, the current work aims at three main objectives:

1. Design a visualization hybrid combustion setup that enables control over oxidizer flux and ignition, while also collecting real-time data, including weight variations from a scale to measure mass fuel consumption.

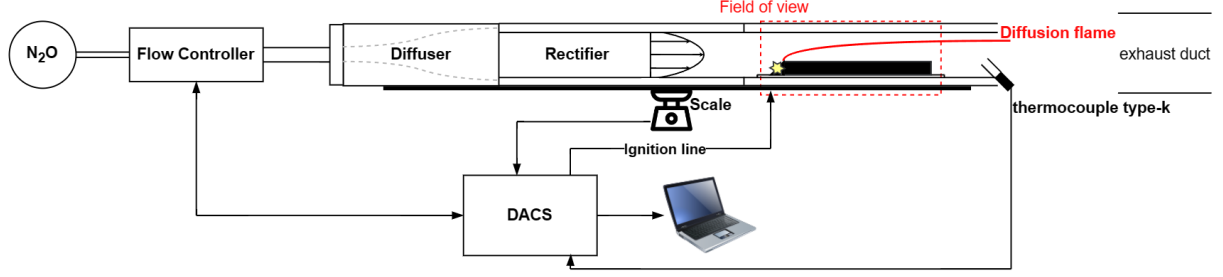


Figure 2: Experimental Setup.

2. Conduct a comprehensive propellant performance analysis, which will include: analyzing the trend behavior of the local regression rate over the fuel length to understand irregularities and to locally estimate the surface combustion of paraffin wax; examining the spatial dependence of the hybrid regression rate law; analyzing the regression rate (mm/s) and the mass fuel consumption (g/s) using two distinct methods based on local and temporal height variation from optical analysis, and temporal mass variations from scale data. Finally, this analysis will compare both methods and their responses to oxidizer flux variations.
3. Analysis of the combustion mechanisms in hybrid motors as described by the hybrid boundary layer combustion theory [3], which includes detecting the outer edge of the thermal boundary layer, locating the flame, and analyzing the flame position within the boundary. Additionally, the present work intends to visualize the liquid layer mechanism responsible for the enhanced regression rate of liquefying fuels.

2. Experimental setup

2.1. Overview

The experimental setup is schematically represented in Figure 2. The system consists of three main components: the feed system, the combustion chamber, and the Data Acquisition and Control System (DACS). The feed system consists of a diffuser and a rectifier that deliver a fully developed oxidizer flow into the chamber.

Once the oxidizer encounters the fuel grain, the user can control the ignition timing, which initiates the combustion process. Following the initial ignition transients, the user controls the timing to increase the oxidizer flux to a previously established level.

The outlet of the flow controller is connected to the rest of the system via a flexible hose, while the remaining rigid parts are mounted on a plate positioned above a scale. From the beginning of the test, the DACS gathers data concurrently from the flow controller, the scale, and a thermocouple

placed downstream. The burning process ceases and the test data is saved once the stop button is pressed, which cuts off the oxidizer feed to the chamber.

The combustion chamber is equipped with two quartz side windows, enabling optical analysis and providing insight into various aspects of hybrid combustion using different experimental imaging techniques.

2.2. Combustion Chamber

The combustion chamber was designed with internal dimensions of 14 x 20 mm. The two side windows are made of 3 mm thickness quartz glass. The chamber is divided into two halves: the first supports the fixed fuel, while the second incorporates the upper region of the chamber. The fuel is fixed to the chamber using a component referred to as "fixator". The fixator was designed to secure fuel slabs up to 10 mm in width and 150 mm in length. Additionally, it covers all sides of the fuel up to a height of 5 mm. The combustion chamber components are shown in Figure 3. The distance between the windows and fuel slab was about 7 mm, this distance was established as a trade-off to minimize thermal stresses and prevent melted wax from splattering onto the windows, while avoiding side burning.

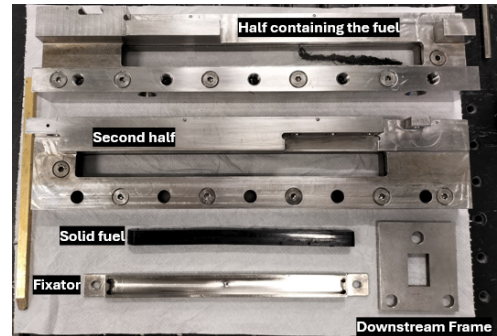


Figure 3: Combustion chamber components.

2.3. Feed system

Nitrous oxide was stored initially on a tank at 44 bar and room temperature. During the test, the

gaseous oxidizer flows from the tank to the flow-controller and then to the main parts of the feed system: the diffuser and rectifier. The diffuser was responsible for a smooth transition from a circular to a 10 x 20 mm inner cross sectional area. This area is similar to the inner port cross-sectional area of the chamber. It has a width similar to the fuel slab, and a minimal height to maximize the oxidizer mass flux entering the chamber, while allowing observation of the combustion processes above the slab. After passing through the diffuser, a flow rectifier was designed with sufficient length to ensure that the turbulent N_2O flow was fully developed across the 10 x 20 mm cross-sectional area before entering the combustion chamber.

2.4. Fuel

For this work, two distinct types of paraffin waxes were used. The waxes were chosen based on their kinetic viscosity to test the inverse dependence of the wax and viscosity as predicted by Karabeyoglu *et al.* [4]. The paraffin, manufactured by Supelco, had a low viscosity (LV) evaluated at 100 °C ranging from 3 to 6 cSt, a low melt point ranging from 46 to 48 °C, and a density ranging from 0.79 to 0.94 g/cm^3 at 15 °C. The other paraffin wax was, type 5910, manufactured by IGI, had a high viscosity (HV) evaluated at 100 °C, ranging from 16.5 to 20.1 cSt, a low melt point ranging from 88.9 to 95.6 °C, and a density of 0.9 g/cm^3 at 15 °C. The first step in producing the final solid fuel slab involved adding 0.5% by mass of carbon black to the neat paraffin (measured with 0.05g precision). The small amount of carbon black was crucial for the optical regression rate measurement tests, facilitating the tracking of the fuel edge during combustion. The blackened paraffin wax (BPW) was melted and well mixed. Then, the BPW was poured into a pre-heated mold and carefully stirred with a toothpick to eliminate any trapped air. The mold sidewalls were then carefully removed and the slab was trimmed to its final dimensions. For the regression rate measurement tests and color image a clear visualization of the fuel inside the chamber was essential, so the slabs had nominal dimension of 10 x 10 x 150 mm. For the other tests, the fuel grain burn was confined to its 2D top surface, shielded along the sides by the fixator walls, the nominal geometry was 10 x 5 x 85 mm. Both geometries have a leading edge with a chamfered shape which was important for an efficient flame ignition.

2.5. DACS

The data acquisition and control system (DACS) includes all the hardware, schematically represented in 5.

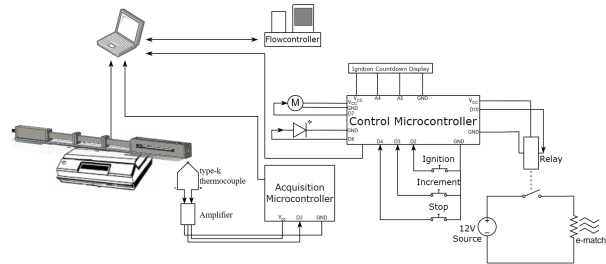


Figure 5: Data Acquisition and Control System.

The main hardware is positioned near the host-computer that operates the main program. After receiving the input of the desired oxidizer mass flow rate for the burn test, the host-computer starts collecting real-time data from multiple devices which include a type-k thermocouple attached downstream for exhaust gases temperature acquisition and a Mettler PE11 scale (0.1g precision) which supports the main rigid parts of the system and was used for regression rate measurements.

After the user inputs stored, the flow controller was adjusted to a mass flow rate of 0.5 Kg/h. This rate was determined as the optimal initial flow rate for igniting the flame without prematurely extinguishing the steel wool burn. When the user pressed the ignition button, a 2-second countdown was displayed on a small screen. Once the countdown finished, the NO relay closed, allowing a 12V battery to discharge to the e-match for 2 seconds. The e-match, attached to the fuel's leading edge, consists of two copper cables with end terminals connected by a Nichrome wire. The copper cables passed through a hole in the lower pre-combustion region of the chamber. Ultra-thin steel wool was wrapped around the Ni-Cr wire and pressed against the fuel surface. When the current flowed through the copper cables, the Ni-Cr wire heated rapidly and ignited the steel wool. The wax was locally heated, leading to flame ignition. The ignition command also triggered a servo motor and a LED, which were used to synchronize the cameras with the main program monotonic time.

Following a 2-second steel wool burn with increased local thermal energy release, the small display stopped showing the message "IGNITION". At this point, the increment button was pressed increasing the flow rate to the previously desired flow rate set during program initiation. After pressing the increment button, combustion with the desired oxidizer mass flow rate occurred within the chamber until the stop button was pressed, setting the flow rate to zero and ending data acquisition process. The collected data was then saved in external file.

The scale operated at a slower baud rate of 2400 bps, and for this reason a minimum interval was

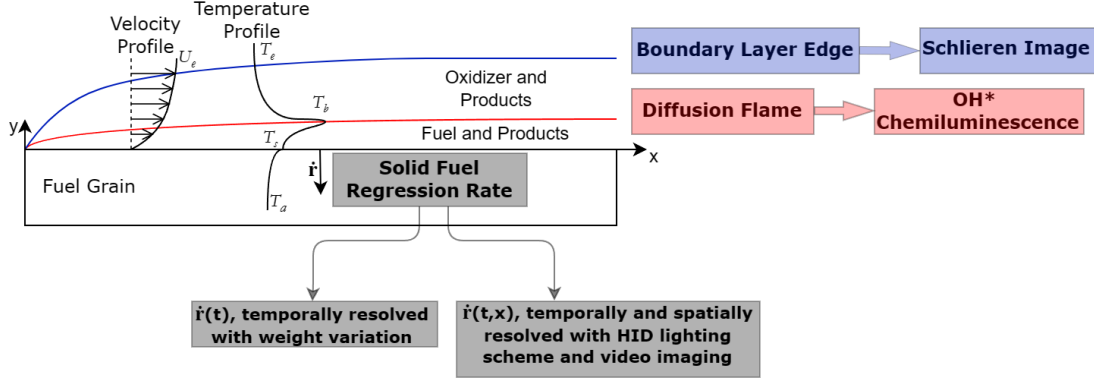


Figure 4: Experimental techniques and corresponding hybrid combustion mechanisms studied.

adjusted to prevent the system from receiving data more frequently than it could handle. Given this, the system collected data at a time resolution of approximately 350 ms. Additionally, the flow controller had an associated uncertainty with a correspondent maximum relative error for the highest measured flow rate of about 1.35%.

2.6. Experimental techniques and Data treatment
The fundamental mechanisms of hybrid rockets and the experimental techniques used for their analysis are shown in Figure 4. In all experimental methodologies, a chessboard calibration pattern was imaged before the tests to estimate absolute positions in millimeters.

The first part of the work involved testing the performance of the propellants and included a setup that allowed for both temporal and spatial resolution of the fuel regression rate. The regression rate was measured via two distinct methods.

The first method used, estimates the regression rate based on the mass fuel consumption during combustion. The fuel mass consumption is given by the mass weight variation collected from the Mettler scale for different instants in time during combustion. Using this data the regression rate can be estimated based on equation 2:

$$\dot{r}_{\text{mass}} = \frac{1}{\rho_f \times A_b} \times \frac{dm}{dt} \quad (2)$$

The regression rate from equation 2 assumes that combustion occurs only along the top surface of the fuel grain, corresponding to a burning area (A_b) of 10 x 150 mm, highlighted in red in the illustrative representation shown in Figure 6. As it neglects the surface area around the sides and back of the fuel grain, this method is expected to slightly overestimate the regression rate of the fuel surface.

To calculate the regression rate, this method employs MATLAB's linear regression model based on the least squares method. A first-degree linear regression line was implemented to quantify the

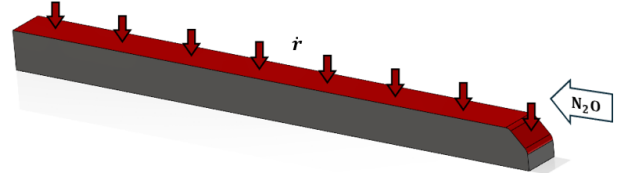


Figure 6: Representation of regression rate using fuel mass consumption.

weight variation. The slope of this line represents the mass fuel consumption (dm/dt).

The second method for estimating the regression rate involved an optical analysis which allowed local measurements across the fuel length during the combustion process. This method is based on the variation of fuel height during combustion as illustrated in Figure 7.

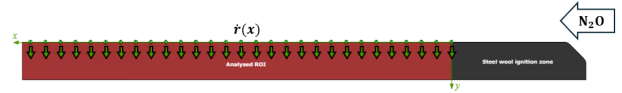


Figure 7: Representation of local regression rate using variation of fuel height.

The regression rate was locally measured after the region near the fuel leading edge with top surface covered in steel wool, used for ignition. The remaining length of the fuel after ignition zone, corresponds to the region of interest (ROI) used for the measurements. In some initial tests the ROI had lower length camera due to slightly narrower camera resolution. In this region, tracking the positions of the fuel edge points during combustion provides the variation in height, which corresponds to the y-displacement variation from the initial fuel edge position. The regression rate using this method follows equation 3:

$$\dot{r} = \frac{dy}{dt} \quad (3)$$

This method assumes that the top wax edge regression rate is uniform across the slab width and combustion process only takes place above the top surface. The setup used for the regression rate measurements using this method is shown in Figure 8.

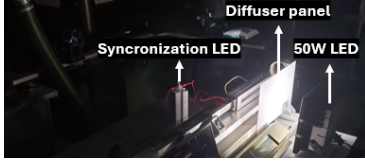


Figure 8: Backlight and direct imaging technique for regression rate optical analysis.

Video frames were captured with a resolution of 2048x701 pixels, using a UV-sensitive sCMOS camera equipped with a 105 mm JENOPTIK UV lens. The frames were acquired at a rate of 100.05 frames per second. However, the actual fps of the video read in MATLAB script for post-processing was between 25 and 30 fps. A 50W LED flood light source was placed behind the combustion chamber, creating a strong contrast between the blackened paraffin wax and the background during the combustion process. This setup provided the strong contrast needed for precise positioning of the top wax edge during post-processing.

The MATLAB post-processing script begins by selecting the first and last frame to be evaluated, followed by the region of interest (ROI) where edge tracking is applied. The initial frame selected corresponds to the first frame after ignition and flux increment. The final frame is chosen as the last one before noise affects edge tracking and produces intense edge displacement fluctuations. For all tests, the ROI starts at the end of the ignition zone covered with steel wool. From here, an edge tracking algorithm is applied on the ROI.

The process begins with the application of Otsu's method to determine the optimal threshold, which effectively segments the saturated and non-saturated regions of the image. This threshold is then used to convert the image into a binary mask, refined by eliminating isolated groups smaller than 50 pixels to reduce noise. Subsequently, the Canny edge detector, using the default MATLAB settings, is applied to precisely delineate the fuel edge that marks the boundary between the two contrasting regions. The positions of the edge points are determined at intervals of 15-pixel columns, with the initial frame setting the time origin and defining the positions of the initial edge points for subsequent measurements of vertical displacement. The script progresses by iterating through all video frames, recording new edge displacements and time of each frame. This process continues until the final frame

is reached, which determines the burn time of the test and captures the last evaluated edge position.

A Gaussian filter with $\sigma=1.1$ is then applied to smooth the displacement measurements, reducing noise and fluctuations within the data. Following this smoothing process, a first-degree linear regression line is fitted to the displacement data using MATLAB's linear regression model, thereby estimating the locally resolved regression rate across all evaluated fuel edge points along the fuel length. Additionally, for an improved trend analysis, the evaluated fuel length was divided into three equal sections.

For all tests, the oxidizer mass flux (G_{Ox}) in $\text{kg}/(\text{m}^2 \cdot \text{s})$ was assumed to be constant during the evaluated burn time and was averaged over this period. The values of the regression rate coefficient a_o in mm/s and flux exponent n in $\text{kg}/(\text{m}^2 \cdot \text{s})$, used in hybrid rockets regression rate law shown in equation 1, were determined using the results from both methods. Logarithms were applied to both sides of the equation followed by first-degree linear fit, the values of a_o and n were then determined. For the locally resolved regression rate data, the fuel length was divided into 1 mm intervals to avoid discrepancies due to differing calibrations between tests. The same linearization procedure was applied to each 1 mm interval, allowing the local estimation of both coefficients.

The time intervals used for sampling data differed between methods. The burn time available for optical analysis was consistently limited, only extending from the moment of flux increase until the fuel height dropped to a level where edge tracking was no longer feasible. On the other hand, during the initial seconds post-flux increment, the scale had a noticeable sensitivity to weight variations before the actual regression behavior could be determined. This issue was considered during the post-processing of the mass results. Since the only experimental variables were the oxidizer flux and the type of paraffin used (LV or HV), both methods are expected to yield comparable regression rate results under the same test conditions, assuming a steady-state regression rate over the burning periods considered.

To compare the two methods, the local regression rate was integrated over the fuel length from the first to the last edge point using the following equation:

$$\frac{dm}{dt} = \rho_f \times \int_{x_1}^{x_N} \dot{r}_{\text{local}} \times w \, dx \quad (4)$$

In equation 4, the width (w) is assumed to be a constant 10 mm along the fuel's length. The variable x represents the length at each integrated edge point from x_1 to x_N where the local regression rate,

was obtained using the optical method. The result of this integration, when multiplied by the density of the solid fuel (ρ_f), yields the mass consumption rate dm/dt . This value is then compared to the weight variation measured by the scale. For this integration, the trapezoidal rule was employed, which assumes a linear variation in the regression rate between two consecutive edge points.

For the flame location and boundary layer height, a similar approach to the used by Jens [7] was employed. A path-integrated image of the reaction zone was obtained using a band-pass filter and collecting light near the OH^* peak emission wavelength, which is around 310 nm. To locate the flame, the peak intensities were identified above each fuel edge point. Schlieren images were also collected, enabling the application of Canny edge detector to estimate the position of the outer edge of the thermal boundary layer. Figure 9 illustrates the Z-type two-mirror Schlieren configuration used alongside the OH^* Chemiluminescence equipment.

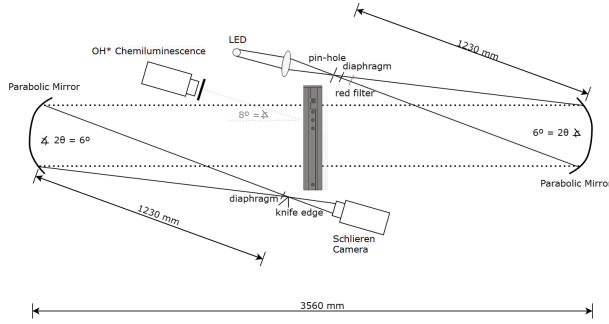


Figure 9: Schlieren setup scheme.

In both cases, the fuel edge coordinates were first identified and then overlaid on the frame under analysis. The evaluated frames were chosen when the flame was well established over the fuel surface and selected for their minimal delay from the flux increment to reduce the effects of the regression rate.

Images of OH^* chemiluminescence were captured at a rate of 2.86 fps, with a resolution of 1296x1201 pixels, using a UV-sensitive sCMOS camera equipped with a 105 mm JENOPTIK UV lens. Positioned after the lens, a Cage Filter Wheel holds an Andover 310FS10-50 band-pass filter, which is centered around 310 nm. In the cage wheel, the holder immediately following the band pass filter was intentionally left empty. After ignition, the cage wheel was turned to the OH^* band-pass filter, capturing the initial light from the LED and synchronizing it with the DACS main program's time. To achieve this synchronization, a servo motor was used for the Schlieren imaging. Schlieren images were recorded with a resolution

of 1024x768 at 30 fps on a JAI CV-M9-GE, 3-CCD RGB camera with a 25 mm Fujinon lens. A red filter was placed after the LED's diaphragm to reduce chromatic aberration and enhance contrast with a longer wavelength penetrating the inhomogeneous media more effectively. A horizontal knife edge was used to enhance the resolution of vertical density gradients along the boundary layer edge. The setup is shown in Figure 10.

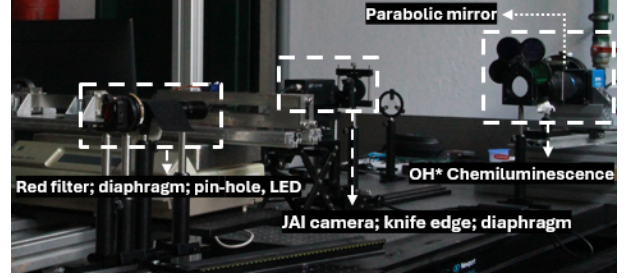


Figure 10: Schlieren setup.

3. Results

3.1. Regression rate measurements

For the regression rate measurement tests, three flux oxidizer levels were tested: $5 \text{ kg}/(\text{m}^2 \cdot \text{s})$, $10 \text{ kg}/(\text{m}^2 \cdot \text{s})$ and $14 \text{ kg}/(\text{m}^2 \cdot \text{s})$.

In the first method three stages of fuel consumption were observed. These stages are evident in Figure 12, which shows the mass fuel variation normalized with the total variation for three levels of flux. Initially, the scale shows sensitivity to the weight variations and the consumption appears slower. It then transitions to a linear behaviour indicative of a steady-state combustion process, and finally, as more fuel is consumed, the consumption rate decrease.

To calculate the regression rate, only the steady-state region with linear behaviour was used, covering the maximum burn time possible for all compared tests, which ranges from 4.109 to 4.5s, as shown in Figure 13. The same procedure was applied to determine the regression rate for the HV BPW tested at higher flux level, as shown in Figure 14.

The results in Figure 13 show that although the dependence on the flux is weak, the regression rate increased with higher oxidizer fluxes. The estimated regression rates, especially for higher fluxes, were consistent with those observed in other slab experiments using GOX [8, 6]. For the same oxidizer flux level, the regression rate estimates for the high-viscosity wax were lower than those for the low-viscosity wax, aligning with the liquefying fuels theory [4].

The spatial resolved regression rate is first analysed in the sections presented in Figure 11. Re-

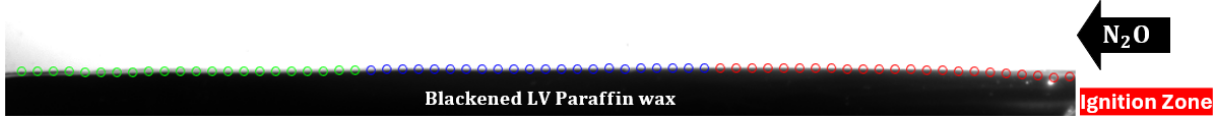


Figure 11: Regions for spatially resolved regression rate trend analysis.

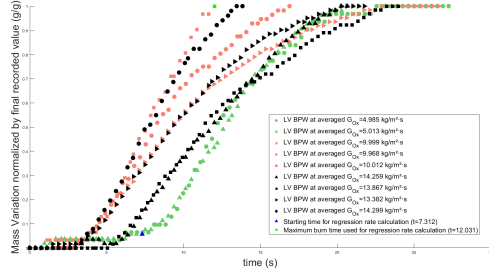


Figure 12: Normalized mass fuel consumption at three different oxidizer flux levels for LV BPW.

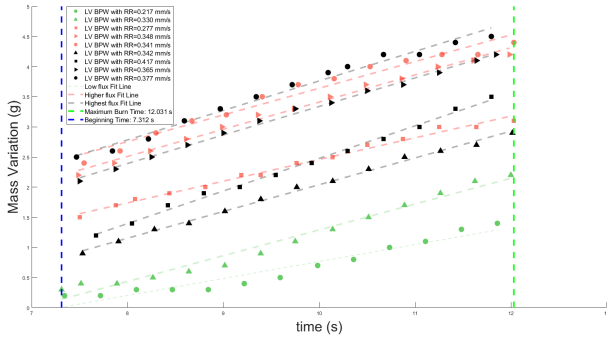


Figure 13: Regression rate calculation for three flux levels during linear consumption behaviour for LV BPW.

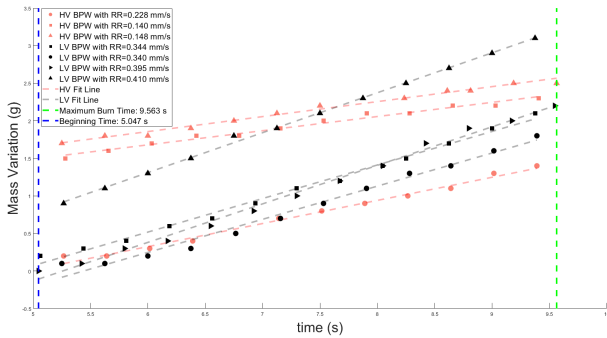


Figure 14: Regression rate calculation during linear consumption behaviour for LV and HV BPW.

sults from two tests at lower flux are shown in Figure 15 and for higher flux in Figure 16. For different flux levels, the displacements revealed a linear variation, indicative of a steady regression rate along the evaluated burn times. This behavior indicates that combustion conditions were stable during the analyzed burn times for the regions examined.

There is a clear distinction between the initial region near the ignition and the rest of the fuel length, this trend is also visualized for the local regression rate results in Figure 17. The regression rate has a steadier behaviour on the middle section indicative of a more uniform combustion and heat transfer to the fuel surface on this region. The ignition zone however is more sensible to stronger temperature gradients and initial irregularities.

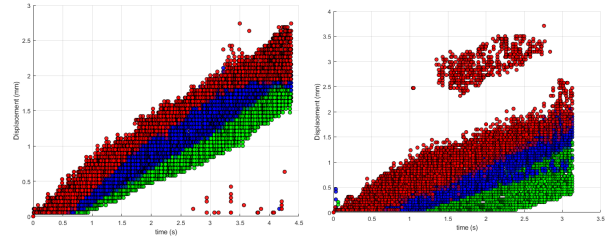


Figure 15: Spatially resolved regression rate for three distinct regions at $5.028 \text{ kg}/(\text{m}^2 \cdot \text{s})$ (left) and $5.085 \text{ kg}/(\text{m}^2 \cdot \text{s})$ (right).

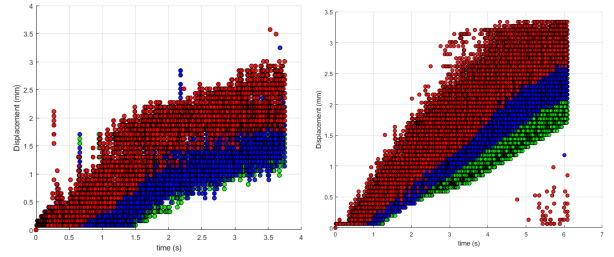


Figure 16: Spatially resolved regression rate for three distinct regions at $10.034 \text{ kg}/(\text{m}^2 \cdot \text{s})$ (left) and $15.011 \text{ kg}/(\text{m}^2 \cdot \text{s})$ (right).

As shown in Figure 17, the regression rate near the ignition zone exhibits higher variation regardless of the flux, indicating a stronger dependence on length in this initial zone, consistent with classic hybrid theory [3]. However, the decrease in regression is not as abrupt as the originally predicted by the term $x^{-0.2}$. In the middle zone, the flux is the dominant variable and the behaviour across the length is steady. The regression rate values estimated overpredict the values in [8, 6] due to greater dependence on the experimental conditions for clear edge tracking and the inclusion of non-steady regions. However, for higher fluxes, the results are closer to those estimated in [9], which also used automated edge tracking, and sub-scaled hybrid motors [2].

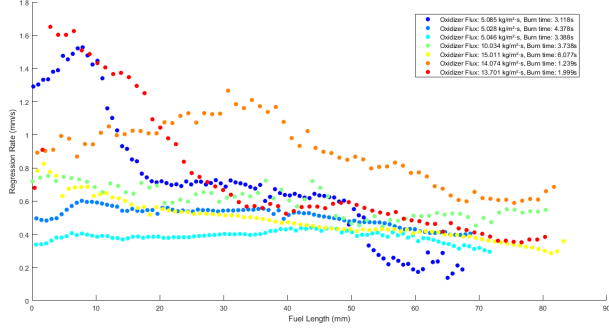


Figure 17: Local regression rate for tests following the most usual trend.

Figure 18 shows the hybrid fuels regression rate law coefficients spatially resolved. The flux exponent (n) values indicate there is a weak dependence on the oxidizer flux for the middle section, which is typical of N_2O . The coefficient a_o is higher than estimated in sub-scale hybrid motor experiments [2], reflecting the trend behaviour of the local regression rate. This value is typically higher than the estimated with paraffin/LOX hybrid motors. However, the optical method overestimates a_o mainly due to the edge tracking method more affected by local burning irregularities. For the mass method, the regression rate law coefficients were $a_o = 0.1561$ and $n = 0.3257$, reinforcing the lower oxidizer dependence of N_2O . The a_o value is very close to the value 0.155 estimated from paraffin/ N_2O sub-scale hybrid motor experiments [2].

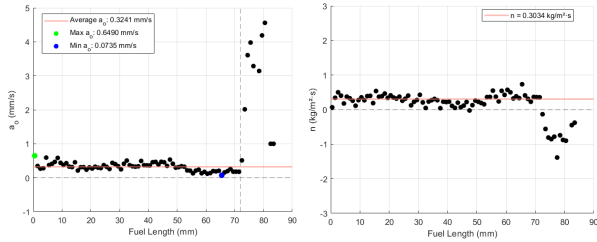


Figure 18: Spatially resolved HRE regression law coefficients a_o on left n on right.

Finally, using equation 4 the results of the mass fuel consumption for all tests, using both methods, are shown in Figure 19. The mass consumption using both methods shows very good alignment for different fluxes. This result demonstrates the reliability of both methods in estimating the overall fuel performance, regardless of the local behaviour, and supports the assumption of a uniform local regression rate across the fuel width. Moreover, the averaged results for different flux levels, shown in the right graph of Figure 19, exhibit a very similar response to the flux variation.

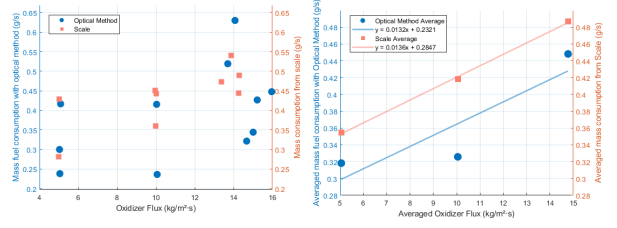


Figure 19: Mass fuel consumption using two distinct methods.

3.2. Flame Location, Boundary Layer Thickness, Liquid Layer

For the flame location and boundary layer growth evaluation, two free-stream fluxes were tested, $10.004 \text{ kg/m}^2 \cdot \text{s}$ and $13.990 \text{ kg/m}^2 \cdot \text{s}$. The results of the flame location are shown in Figure 20. The lower position with higher flux can be explained by enhanced convective heat transfer to the fuel surface, resulting in more vaporization of fuel which feeds the reaction zone, maintaining the flame closer to the fuel surface. For the thermal boundary layer results, shown in Figure 21, the higher flux with higher heat transfer and thermal gradients explains the lower boundary layer for higher flux. At further distances from leading edge, no differences were found between boundary layers due to low flux regime and atmospheric pressure conditions. For the results of the flame within boundary layer shown in Figure 22, frames at times close as possible were used and a free-stream flux of $13.990 \text{ kg/m}^2 \cdot \text{s}$. At the evaluated positions of 10, 20 and 30 mm from fuel leading edge, the position of the flame is consistently higher than predicted in classic fuels by Marxman *et al.* [3], as estimated in [7]. Also, peak signals extend more broadly across the length for $x = 30 \text{ mm}$, indicating a thickening of flame region for further distance. These results are in agreement with [7]. At greater distances, due to the experimental conditions with low oxidizer flux regime, the flame exhibited steadier behavior and did not follow the high boundary layer growth.

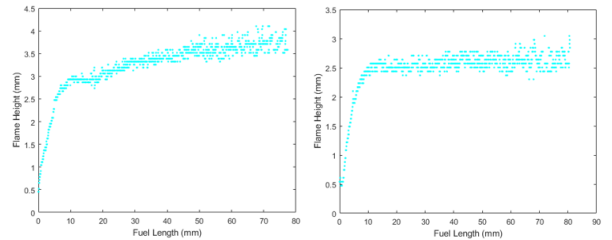


Figure 20: Flame height estimation with free-stream at $10.004 \text{ kg/(m}^2 \cdot \text{s)}$ (left) and $13.990 \text{ kg/(m}^2 \cdot \text{s)}$ (right).

Additionally, a color image is presented in Figure 23 showing the liquid layer mechanism [4].

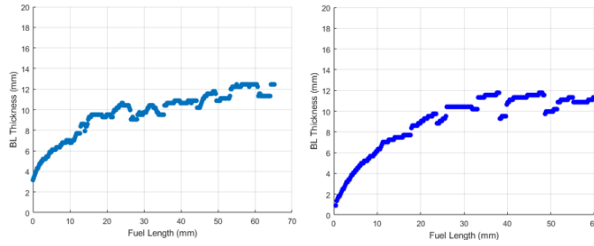


Figure 21: Boundary Layer Thickness estimation with free-stream at $10.004 \text{ kg}/(\text{m}^2 \cdot \text{s})$ (left) and $13.990 \text{ kg}/(\text{m}^2 \cdot \text{s})$ (right).

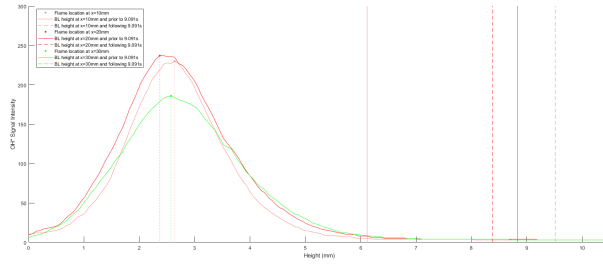


Figure 22: OH* Signal distribution and height positioning of flame and boundary layer. Frames time from ignition.



Figure 23: Liquid Layer forming on fuel surface.

4. Conclusions

A comprehensive analysis of the combustion of blackened paraffin wax with N_2O at low flux regime and atmospheric pressure was conducted. The regression rate results using the mass method showed similar results with small-scale hybrid motors and other slab burner experiments. The zone with higher length dependency near the ignition zone was found to decrease in length not as abruptly as predicted by classic theory. The middle zone exhibited steadier behaviour and more flux-dependent. The hybrid fuels regression rate law showed bigger dependency on a_o , reflecting the local trend and low oxidizer flux typical of N_2O . The methods were proven to align well and accurately estimate overall fuel performance based on mass fuel consumption, also showing similar response to the oxidizer flux. Moreover, the flame and boundary layer height reduced with increased flux due to higher heat transfer and thermal gradients. The flame position within the boundary layer was found to be higher than that of classic fuels to some extent. Thickening of the fuel zone at further length positions was observed. Finally, the liquid layer responsible for enhanced mass transfer and consequently

increased regression rate was visualized.

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